

1 TITLE SLIDE

- Welcome to my PhD defense on bolometry diagnostics and real-time radiation feedback control at Wendelstein 7-X
- This work addresses two critical challenges for future fusion reactors: (1) protecting plasma-facing components through controlled radiation, and (2) achieving stable detachment at high radiation fractions ($f_{\text{rad}} \geq 85\%$)
- Three main contributions: implemented first bolometer-based real-time feedback at W7-X, optimized channel selection through systematic sensitivity analysis, and validated MFR tomography with radially dependent anisotropy weighting

2 CONTENTS

- Presentation structure: (1) Motivation - fusion energy and W7-X challenges, (2) Bolometer diagnostic system - 94 channels with ~ 5 cm spatial resolution, (3) Real-time feedback - achieved $f_{\text{rad}} \geq 85\%$ with factor ≥ 2 heat load reduction, (4) LOS sensitivity - identified optimal channel configurations, (5) Tomography - MFR with RDA for 2D radiation reconstruction, (6) Conclusions and future work

3 MOTIVATION

3.1 Nuclear Fusion: Energy Challenge

- Nuclear fusion through D-T reactions offers clean, abundant energy - deuterium from seawater, tritium bred from lithium
- Lawson criterion: $n_e \cdot \tau_E \cdot T \geq 3 \times 10^{21} \text{ keV}\cdot\text{s}/\text{m}^3$ defines conditions for net energy gain ($Q > 1$)
- Requires extreme conditions: $T \sim 10\text{--}20 \text{ keV}$ (100–200 million °C) and confinement time $\tau_E \sim 1\text{--}10 \text{ s}$
- Magnetic confinement uses strong fields (2.5 T at W7-X) to contain hot plasma away from material walls
- Stellarators like W7-X achieve steady-state operation through 3D-shaped coils, unlike pulsed tokamaks

3.2 Wendelstein 7-X

- W7-X is the world's largest and most advanced stellarator: $R = 5.5 \text{ m}$, $a = 0.53 \text{ m}$, $B = 2.5 \text{ T}$, optimized for steady-state operation
- Key challenge: managing heat loads on plasma-facing components during long pulses (up to 30 minutes planned)
- Island divertor concept: 5 helical island chains intercept heat flux, connection lengths 100–1000 m (10–100× longer than tokamaks)
- Detachment strategy: radiate power in scrape-off layer and edge to reduce divertor heat loads from $\sim 10 \text{ MW}/\text{m}^2$ to $< 5 \text{ MW}/\text{m}^2$
- Goal: achieve stable detachment at $f_{\text{rad}} \geq 85\text{--}90\%$ without plasma disruption while maintaining core performance
- This requires real-time control of radiation distribution - motivation for bolometer feedback system

3.3 Plasma Radiation

- Plasma radiation mechanisms: (1) Bremsstrahlung (free-free transitions, $\sim T_e^{1/2}$), (2) Line radiation (bound-bound, temperature-dependent), (3) Recombination (free-bound)
- Impurity seeding strategy: inject low-Z (N_2) or high-Z (Ne, Ar) gases to enhance edge radiation while avoiding core contamination
- Carbon and oxygen are intrinsic impurities from wall interactions - carbon dominates by factor $10^2 \times$ over oxygen at W7-X

- Radiation cooling function $P_{\text{rad}}/(V \cdot n_e \cdot n_{\text{imp}})$ peaks at different temperatures for each impurity: C peaks $\sim 10\text{--}50\text{ eV}$ (edge), O peaks $\sim 100\text{ eV}$
- Transport in W7-X: neoclassical baseline with anomalous contributions from turbulence (ITG, TEM modes)
- Anomalous transport increases diffusion by factor 10–100 over neoclassical, critical for impurity transport modeling
- Radiation fraction $f_{\text{rad}} = P_{\text{rad}}/P_{\text{ECRH}}$ must be balanced: $f_{\text{rad}} < 70\%$ insufficient for divertor protection, $f_{\text{rad}} > 95\%$ risks core cooling and confinement degradation
- Optimal operating window: $f_{\text{rad}} = 85\text{--}90\%$ achieves detachment while maintaining plasma stability

4 CORE BOLOMETER AT W7-X

4.1 Bolometer Diagnostic (Hardware)

- Bolometers measure total plasma radiation power (broadband, 0.2–600 nm) through resistive heating of metal absorbers
- Metal resistor design: thin gold film ($1.5\text{ }\mu\text{m}$) on mica substrate, absorbs radiation $\rightarrow$ temperature increase $\rightarrow$ resistance change
- Wheatstone bridge configuration: AC excitation (5 V, kHz) eliminates 1/f noise, differential measurement doubles signal
- Three camera arrays provide comprehensive plasma coverage: HBC (32 channels, horizontal), VBCl/r (2×31 channels, vertical)
- Total 94 channels installed, 61 functional in OP1.2b campaign (some damaged by ECRH stray radiation)
- Each channel integrates radiation along line of sight - use geometry matrix $\mathbf{T}$ and etendue K_M to reconstruct 2D distribution
- Spatial resolution $\sim 5\text{ cm}$ at magnetic axis, determined by etendue width and viewing geometry

4.2 Performance

- Calibration is critical for absolute power measurements - three methods used: (1) laser, (2) electrical ohmic heating, (3) in-situ during operation
- In-situ ohmic heating calibration: apply $U_{\text{cal}} = 1.2\text{--}2.5\text{ V}$ for 1.6 s, measure current evolution $I(t)$, extract R_M , τ_M , κ_M from exponential decay
- Calibration parameters (OP1.2b median): $R_M = 0.987 \pm 0.011\text{ }\Omega$, $\kappa_M = 0.724 \pm 0.055\text{ A}^2$, $\tau_M = 110.57 \pm 4.93\text{ ms}$, remarkably stable across 1182 experiments
- Etendue K_M quantifies geometric light collection efficiency: accounts for detector-aperture geometry, partial shadowing, angle-dependent transmission
- Time resolution configurable: $\{0.4, 0.8, 1.6, 3.2, 6.4, 12.8\}$ ms sampling available, 1.6 ms used for feedback (balances speed vs noise)
- Performance metrics: $\text{SNR} > 1000$ at high radiation, noise $\sigma_{\Delta U} = 0.339 \pm 12.586\text{ }\mu\text{V}$, Gaussian error $1.6\text{ }\mu\text{W}$, drift $0.057 \pm 27.621\text{ }\mu\text{V/s}$
- Chord brightness $P_{\text{ch}} = P_M/K_M$ provides radial profile information - shows radiation distribution from core to SOL
- Absorption efficiency near unity ($> 95\%$) across 0.2–600 nm wavelength range - truly broadband measurement

5 REAL-TIME RADIATION FEEDBACK CONTROL

5.1 Feedback System (Island Divertor)

- W7-X island divertor concept: 5 helical island chains ($m/n = 5/5$) at plasma edge intercept heat and particle flux

- Each island chain has upper and lower divertor modules - 10 modules total arranged helically around torus
- Connection lengths $L_c = 100\text{--}1000\text{ m}$ ($10\text{--}100\times$ longer than tokamaks) allow extensive radiation cooling along field lines
- Standard magnetic configuration: $\iota/2\pi = 1$ (rotational transform), creates 5-fold symmetric island structure
- Gas injection system: 10 piezo valves for fast response ($< 10\text{ ms}$), thermal helium beam valves for feedback control
- Valve locations strategically placed: HM1-5 modules (upper/lower pairs) allow targeted seeding into specific island chains
- Island X-points are radiation hot spots - field lines converge, increasing local emissivity by factor $2\text{--}5\times$
- Bolometer cameras positioned to view these X-points - critical for feedback sensitivity
- Field line tracing shows direct connection between gas injection points and divertor targets via island chains

5.2 Feedback System (Implementation)

- First bolometer-based real-time radiation feedback system at W7-X, operational during OP1.2b campaign (2018)
- Hardware: NI 6321 DAQ card for fast acquisition, dedicated analog outputs for PID controller, parallel processing architecture
- Two prediction proxies implemented: (1) $P_{\text{pred}}^{(1)}$ = geometry-weighted sum over selected channels S , (2) $P_{\text{pred}}^{(2)}$ = single channel voltage (dimensionless, minimal overhead)
- Channel selection S optimized through systematic analysis: VBC subset $S = \{61, 65, 73, 78, 86\}$ viewing separatrix and X-points achieves $\geq 85\%$ accuracy
- Alternative HBC selections also viable: $S = \{4, 7, 8, 20, 23, 27\}$ provides similar performance with different viewing geometry
- PID controller parameters: $K_p = 0.5\text{--}2.0$ (proportional gain), $K_i = 0.1\text{--}0.5$ (integral), $K_d = 0$ (derivative not needed due to slow plasma response)
- Setpoint: f_{rad} target (typically $85\text{--}90\%$), controller adjusts gas valve opening to maintain radiation fraction

5.3 Feedback System (Latency)

- System latency breakdown: acquisition 13.6 ms (minimum, validated with laser diode), FIFO smoothing $\sim 8\text{ ms}$ ($M = 10$ samples $\times 1.6\text{ ms}$), processing $< 1\text{ ms}$
- Total feedback loop latency: $150\text{--}400\text{ ms}$ dominated by plasma response time (gas transport, ionization, radiation buildup)
- FIFO moving average filter: reduces noise by factor $\sqrt{M}$ but adds latency $M \cdot \Delta t/2 \approx 8\text{ ms}$ for $M = 10$
- Trade-off: faster sampling ($\Delta t = 0.8\text{ ms}$) reduces latency but increases noise, 1.6 ms optimal for feedback application
- Benchmark test with laser diode: confirmed 13.6 ms acquisition latency, validated timing chain from photon to digital signal
- Future improvements: reduce acquisition to $< 10\text{ ms}$ through hardware upgrades, implement predictive algorithms to compensate for plasma response delay
- Comparison: Thomson scattering $\sim 50\text{ ms}$, interferometry $\sim 1\text{ ms}$, but bolometry measures the actual control variable (radiation)

5.4 Experimental Achievements (XP20181010.32)

- XP20181010.32 - breakthrough experiment achieving stable feedback-controlled detachment with hydrogen seeding
- Discharge parameters: duration 9.2 s, $P_{\text{ECRH}} = 6.23$ MW, standard magnetic configuration ($\iota/2\pi = 1$), total input energy 55 MJ
- Key achievement: sustained $f_{\text{rad}} \geq 85\%$ with peaks reaching 100% ($P_{\text{rad}} = P_{\text{ECRH}}$), no plasma disruption or termination
- Divertor heat load reduction: factor $\geq 2\times$ from baseline ~ 10 MW/m² to < 5 MW/m², validated by infrared thermography
- Detachment signatures: C²⁺ filterscope emission appears at $f_{\text{rad}} \sim 50\%$, intensifies and moves poloidally as f_{rad} increases
- Prediction accuracy: $P_{\text{pred}}^{(1)}$ tracks P_{rad} within 10–15% throughout discharge, validates 5-channel VBC subset $S = \{61, 65, 73, 78, 86\}$
- Plasma stored energy W_{dia} maintained at ~ 0.8 – 1.0 MJ despite high radiation - demonstrates compatibility with good confinement
- This demonstrates reactor-relevant scenario: high f_{rad} for component protection while maintaining plasma performance

5.5 Experimental Achievements (Discussion)

- Bolometer feedback advantages: (1) directly measures control variable (total radiation), (2) broadband - insensitive to impurity mix, (3) fast response (1.6 ms sampling)
- Comparison with alternatives: density feedback (DI) is indirect - affects confinement, can destabilize; C-III filterscope measures only one charge state, misses O, other C ions
- Bolometer robustness: total radiation measurement independent of impurity species composition (C vs O vs seeded gases)
- Spatial information: multiple viewing chords reveal radiation distribution - core vs edge vs SOL, identify X-point hot spots
- Physical connection: radiation IS the detachment mechanism - controlling P_{rad} directly controls heat dissipation
- Gas injection optimization: moderate puff duration (50–200 ms) optimal - short pulses cause transients, long pulses ineffective
- PID tuning challenges: parameters discharge-specific, depend on magnetic configuration, heating power, impurity species
- No universal scaling law found - complex parameter space requires adaptive control strategies
- System limitations: 150–400 ms total latency (dominated by plasma response), limits achievable control bandwidth to ~ 2 – 5 Hz
- Future improvements: (1) reduce acquisition latency to < 10 ms, (2) implement predictive control using plasma models, (3) machine learning for automatic PID optimization
- Despite limitations, bolometer feedback is the most direct method for radiation control in reactor-relevant scenarios
- Validated approach: successfully demonstrated at W7-X, applicable to ITER, DEMO, and future stellarators

6 LINE-OF-SIGHT SENSITIVITY

6.1 LOS Sensitivity (Methodology)

- Systematic LOS sensitivity analysis to identify optimal channel configurations for feedback control
- Dataset: 45 discharge pairs from OP1.2b campaign spanning wide parameter range ($P_{\text{ECRH}} = 3.5$ – 6.5 MW, $f_{\text{rad}} = 0.3$ – 1.0 , $T_e = 0.9$ – 1.5 keV)

- Methodology: evaluate all possible channel combinations ($C(32, m)$ for HBC, $C(62, m)$ for VBC) for $m = 3, 5, 7$ channels
- Six evaluation metrics applied: (1) correlation, (2) weighted deviation, (3) mean deviation, (4) Fourier transform correlation, (5) Fisher information, (6) χ^2 fitness
- Each metric quantifies how well P_{pred} from channel subset S predicts total P_{rad}

6.2 LOS Sensitivity (Results)

- Key finding: channels viewing separatrix ($\rho_{\text{pol}} \sim 0.95\text{--}1.05$) and SOL most effective - this is where detachment radiation concentrates
- VBC cameras outperform HBC for edge sensitivity: better viewing geometry toward X-points and island chains
- Global maximum in sensitivity profiles: VBC channel viewing upper inboard X-point ($r_{\text{eff}} \approx -0.13 r_a$)
- HBC shows local maxima at $\pm 0.4 r_a$, $\pm 0.7 r_a$ corresponding to island chain intersections
- Robustness: multiple channel combinations achieve $\geq 85\%$ prediction accuracy - no single “best” set exists
- Optimal channel count: $m = 3\text{--}5$ sufficient, $m = 7$ provides marginal improvement ($< 5\%$) at cost of increased complexity
- Weighted deviation threshold: $\Delta_w < 0.15$ defines acceptable performance (accounts for measurement uncertainties and temporal variations)
- Correlation with plasma parameters: high $P_{\text{ECRH}} (\geq 4.2 \text{ MW})$, high $f_{\text{rad}} (> 0.75)$, $T_e \sim 1.3 \text{ keV}$ yield best prediction quality

7 TOMOGRAPHY

7.1 Minimum Fisher Regularization (Theory)

- Tomography problem: invert line-integrated measurements ($\mathbf{T}\mathbf{x} = \mathbf{b}$) to reconstruct 2D emissivity distribution $\mathbf{x}$
- Ill-posed inverse problem: 61 measurements but ~ 4500 pixels ($30 \text{ radial} \times 150 \text{ poloidal}$ grid) - severely underdetermined
- Geometry matrix $\mathbf{T}$: encodes etendue-weighted line integrals, condition number $\sim 10^6$ indicates strong ill-posedness
- Regularization required: Minimum Fisher Regularization (MFR) with Tikhonov-type penalty minimizes $\|\mathbf{T}\mathbf{x} - \mathbf{b}\|^2 + \lambda^{-1} \|\mathbf{H}\mathbf{x}\|^2$
- Fisher information $I_F = \int (1/x) (\partial x / \partial r)^2 dr$ measures gradient content - lower I_F = smoother solution
- Regularization matrix $\mathbf{H} \propto (1/x) \nabla^T \nabla$ penalizes large gradients weighted by local emissivity
- Radially dependent anisotropy (RDA): different smoothness in radial vs poloidal directions, $k_{\text{ani}}(r)$ varies with radius
- Core ($k_{\text{ani}} \approx 2\text{--}10$): favors smooth, flux-surface aligned emission; Edge ($k_{\text{ani}} \approx 0.1\text{--}0.5$): allows localized, anisotropic structures
- Iterative solution: converges in $N_T = 10\text{--}20$ iterations, regularization parameter λ chosen by L-curve method
- Computational grid: 30×150 cells (radial $\times$ poloidal), domain extends to $1.3^2 V_P$ to handle boundary radiation

7.2 Minimum Fisher Regularization (RDA)

- RDA implementation: $k_{\text{ani}}(r)$ profile transitions smoothly from core to edge using tanh function
- Typical profiles: $k_{\text{ani}} = \{2.0, 0.3\}$ (core, edge) for standard cases, $\{5.0, 0.5\}$ for highly anisotropic phantoms

- Physical motivation: core radiation follows flux surfaces (isotropic in flux coordinates), edge shows X-point localization (anisotropic)
- Advantages of MFR+RDA: (1) incorporates physics knowledge, (2) robust to noise, (3) stable convergence, (4) no negative emissivities
- Alternative: Relative Gradient Smoothing (RGS) can be combined with RDA for enhanced feature localization
- RGS weighting: $\mathbf{H}_{\text{RGS}} \propto (\nabla x/x)^2$ emphasizes relative gradients, stronger localization but may sacrifice global accuracy
- Comparison: RDA chosen as primary method - better balance between accuracy and robustness for W7-X geometry

7.3 Benchmarking (Geometry)

- Geometry perturbation study: tested camera position variations ($\pm 2^\circ$ tilts), detector/aperture segmentation ($N = 2, 4, 8$), artificial symmetric geometries
- Segmentation convergence: $N \geq 8$ provides adequate resolution (etendue variations $< 10^{-11} \text{ mm}^{-3}$), $N = 8$ chosen for computational efficiency
- Camera tilt sensitivity: $\pm 2^\circ$ rotations cause etendue changes $\sim 10^{-11} \text{ mm}^{-3}$, forward model variations 0.5–1% in chord brightness
- Intrinsic asymmetry: HBC 68.75° tilt causes 0.5–1% left-right asymmetry in forward calculations even with symmetric radiation
- Artificial symmetric HBC: mirroring half the array reduces asymmetry, validates that as-designed geometry is near-optimal
- Worst-case scenario: $\pm 2^\circ$ misalignment could shift reconstructed features by $\sim 0.2 r_a$, emphasizes need for accurate as-built measurements
- Etendue validation: numerical integration agrees with analytical solutions for simple geometries (parallel plates, point sources)
- Conclusion: discretization computationally stable, geometry uncertainties non-negligible but manageable with careful calibration

7.4 Phantom Reconstructions

- Phantom benchmark suite: 12+ synthetic radiation distributions testing algorithm performance across parameter space
- Test cases include: (1) simple rings at various radii, (2) Gaussian features, (3) anisotropic island-like structures, (4) nested concentric profiles, (5) asymmetric distributions
- Noise robustness: tested with up to 10% Gaussian noise on measurements, algorithm maintains $\rho_c > 0.85$ and $\chi^2 < 2$
- Parameter variations: systematic scans of k_{ani} (0.1–20), radial position (0.3– $1.3 r_a$), intensity (0.1–2 MW/m³), width (σ_r, σ_θ)
- Quality metrics: (1) correlation ρ_c (phantom vs tomogram), (2) χ^2 fitness (forward vs measured), (3) MSD (mean squared deviation), (4) integrated powers $P_{\text{rad}, 2D}$
- Key findings: P_{rad} underestimates P_{2D} by $\sim P_{2D}^{(\text{SOL})}$ (standard extrapolation misses SOL contribution), χ^2 and ρ_c not always congruent
- Optimal parameters identified: $k_{\text{ani}} = \{2.0, 0.3\}$ for standard cases, $\{5.0, 0.5\}$ for highly anisotropic, $N_T = 15$ iterations, grid 30×150
- Reconstruction quality: $> 90\%$ for well-constrained phantoms (good camera coverage), 70–85% for challenging cases (localized features, poor viewing angles)
- Recommended settings validated through 59 phantom tests with comprehensive parameter variations

7.5 Experimental Reconstructions

- Experimental reconstructions: applied MFR+RDA to 4 feedback-controlled discharges from OP1.2b (XP20180809.13, XP20181010.32, XP20181011.12, XP20181018.45)
- Temporal evolution: reconstructions show radiation moving from core to edge as f_{rad} increases, validates detachment progression
- Spatial features resolved: X-point localization (factor $2\text{--}3\times$ local enhancement), island chain structure, MARFE-like asymmetries
- XP20180809.13 example: laser blow-off events clearly visible as transient core radiation spikes, demonstrates temporal resolution
- Power balance application: $P_{2D}^{(\text{core})} + P_{2D}^{(\text{SOL})}$ provides alternative to standard P_{rad} extrapolation
- Agreement: $P_{2D}^{(\text{total})}$ within 10–20% of P_{rad} , discrepancy attributed to geometry uncertainties and SOL extrapolation errors
- Key finding: standard extrapolation systematically underestimates total power by missing SOL contribution ($\sim 15\text{--}25\%$)
- Core radiation $P_{2D}^{(\text{core})}$ more stable metric for quasi-steady-state analysis - less sensitive to edge fluctuations
- Statistical analysis (4 discharges, 50+ time slices): $P_{\text{rad}} \propto P_{2D}^{(X)}$ correlation strong ($R^2 > 0.85$), validates reconstruction accuracy
- Quality metrics: $\chi^2 = 1.2\text{--}2.5$ (acceptable), $\rho_c = 0.75\text{--}0.90$ (good correlation), $\text{MSD} < 0.3$ (low deviation)
- Discrepancy between χ^2 and ρ_c : optimal tomogram may not have $\chi^2 = 1$ due to systematic errors (geometry, calibration)
- Anisotropy optimization remains challenging: high-dimensional parameter space (k_{core} , k_{edge} , transition width), profile-dependent
- Computational performance: $\sim 2\text{--}5$ seconds per frame on standard CPU, suitable for post-shot analysis but not real-time
- Future prospects: GPU acceleration could reduce to < 100 ms per frame, enabling real-time tomography for advanced feedback

8 CONCLUSIONS

8.1 Key Achievements

- Three major achievements form integrated diagnostic and control system for W7-X radiation management
- (1) Real-time feedback system: first bolometer-based radiation control at W7-X, achieved $f_{\text{rad}} \geq 85\%$ with factor ≥ 2 heat load reduction
- (2) LOS optimization: systematic analysis of 45 discharge pairs, identified robust channel configurations ($m = 3\text{--}5$ sufficient)
- (3) Tomography: validated MFR+RDA algorithm through 59 phantom tests and 4 experimental reconstructions
- Integration: feedback uses optimized channels, tomography provides detailed 2D validation of radiation distribution
- Reactor relevance: demonstrated stable high-radiation scenarios essential for ITER, DEMO divertor protection

Comprehensive Summary:

- Feedback system specifications: minimum acquisition latency 13.6 ms, total loop 150–400 ms (plasma-limited), PID control with dual prediction proxies
- Benchmark discharge XP20181010.32: 9.2s duration, $P_{\text{ECRH}} = 6.23$ MW, $f_{\text{rad}} \geq 85\%$ sustained, heat load reduction factor ≥ 2 , no disruption

- Diagnostic characterization: 94 channels installed (61 functional OP1.2b), calibration stability $< 5\%$ over 1182 experiments, $\text{SNR} > 1000$
- LOS sensitivity results: separatrix/SOL viewing channels optimal, VBC outperforms HBC for edge, multiple viable configurations identified
- Detachment physics insights: C^{2+} signature at $f_{\text{rad}} \sim 50\%$, full detachment $f_{\text{rad}} > 90\%$, radiation shifts inward to $\rho_{\text{pol}} \approx 0.95$
- STRAHL modeling: carbon dominates by $10^2\times$ over oxygen, $\text{C}^{3+}/\text{C}^{2+}$ critical near LCFS, validates feedback channel selection
- Tomography performance: reconstruction quality $\rho_c > 0.85$, $\chi^2 < 2.5$, P_{2D} within 10–20% of P_{rad} , resolves X-point structures
- Power balance improvement: $P_{2D}^{(\text{core})}$ provides stable metric, reveals 15–25% SOL underestimation in standard extrapolation
- All methods extensively validated through OP1.2b campaign (2018): 1182 experiments, 45 feedback discharges analyzed

9 OUTLOOK

9.1 Future and Possibilities

- System upgrades roadmap: (1) hardware - reduce latency to $< 10\text{ ms}$ through faster DAQ, (2) software - implement predictive control algorithms
- Machine learning applications: (1) automatic PID optimization for different scenarios, (2) optimal channel selection, (3) predictive gas injection timing
- Advanced modeling: integrate STRAHL with EMC3-EIRENE for 3D impurity transport, validate against OP2.1 high-performance scenarios
- Improved chamber models: refine two/three-chamber parameters with OP2.1 data, develop predictive scaling laws for gas injection
- Tomography extensions: (1) expand phantom benchmark suite to > 50 cases, (2) GPU acceleration for real-time capability ($< 100\text{ ms/frame}$)
- Real-time tomography vision: 2D radiation distribution feedback would enable spatial control (target specific regions), detect MARFEs early
- Automated parameter optimization: Bayesian methods or grid search to find optimal k_{ani} profiles for different radiation scenarios
- ITER/DEMO relevance: methods directly transferable, DEMO requirements ($P_{\text{fus}} \sim 500\text{ MW}$, $f_{\text{rad}} \geq 95\%$, $q_{\text{div}} < 10\text{ MW/m}^2$) demand reliable radiation control
- Stellarator applicability: techniques applicable to other stellarators (LHD, future devices), tokamaks (ITER, DEMO, SPARC)
- Long-term vision: fully automated radiation control system with adaptive learning, spatial distribution optimization, predictive capabilities

10 CLOSING

10.1 Thank You

- Thank you for your attention - “Sin é, a chairde” (That’s it, friends) - Irish Gaelic closing
- Open for questions on any aspect: feedback system implementation, LOS optimization methodology, tomography algorithm details, experimental results interpretation
- Key discussion points prepared: (1) latency reduction strategies, (2) comparison with alternative feedback methods, (3) tomography parameter optimization, (4) ITER/DEMO applicability
- Acknowledge W7-X team, IPP collaborators, EUROfusion support, Helmholtz Association, University of Greifswald

BACKUP SLIDES - Quick Reference

Plasma Physics:

- Transport mechanisms: classical, neoclassical, anomalous
- Impurity transport: $D_{\text{an}} \sim 0.3\text{--}3\text{ m}^2/\text{s}$ dominates
- Detachment: $T_e < 5\text{ eV}$ threshold, $P_{\text{div}} \rightarrow 0$

Modeling & Analysis:

- Two/three-chamber seeding models
- Power balance: $P_{\text{bal}} \approx 0$ within errors

Diagnostic Details:

- Etendue: $\tilde{K}_M$ quantifies light collection
- Calibration: R_M, τ_M, κ_M from ohmic heating
- Performance: $\text{SNR} > 1000$, noise $0.339\text{ }\mu\text{V}$
- Bolometer equation: AC bridge, power balance

STRAHL Modeling:

- 1D impurity transport code
- Carbon dominance: $10^2\times$ over oxygen
- Critical charge states: $\text{C}^{3+}/\text{C}^{2+}$ near LCFS
- Radiation shift with f_{rad}

LOS Sensitivity Details:

- Six metrics: correlation, deviation, FFT, Fisher, χ^2
- Geometry sensitivity: $\pm 2^\circ$ tilt acceptable
- Segmentation: $N=8$ optimal
- RDA vs RGS: RDA more robust
- Plasma parameters: $P_{\text{ECRH}} \geq 4.2\text{ MW}$ optimal

DAQ Latency:

- Laser characterization: 13.6 ms minimum
- Sample times: 0.8–3.2 ms tested
- Total loop: 150–400 ms sufficient